

Module 1: Network + Network Fundamentals and building networks

Day 1

Task 1:

Time	Source	Destination	Protocol	Length	Info
2654	10.769681	192.168.1.5	TCP	66	51998 → 443 [ACK] Seq=518 Ack=3798 Win=128640
2655	10.774887	192.168.1.5	TLSv1	130	Change Cipher Spec, Application Data
2656	10.774562	192.168.1.5	TLSv1	1300	Application Data
2659	10.790108	3.110.26.57	TLSv1	145	Application Data
2660	10.790109	3.110.26.57	TLSv1	145	Application Data
2661	10.790219	192.168.1.5	TCP	66	51998 → 443 [ACK] Seq=1816 Ack=3956 Win=138800
2667	10.817661	3.110.26.57	TCP	1506	443 → 51998 [ACK] Seq=3956 Ack=1816 Win=88896
2668	10.817663	3.110.26.57	TCP	1506	443 → 51998 [PSH, ACK] Seq=5396 Ack=1816 Win=800
2669	10.817664	3.110.26.57	TLSv1	1447	Application Data
2670	10.817665	3.110.26.57	TLSv1	93	Application Data
2672	10.817897	192.168.1.5	TCP	66	51998 → 443 [ACK] Seq=1816 Ack=6244 Win=126704
2900	11.648586	17.242.13.4	TLSv1	421	Application Data
2901	11.648706	192.168.1.5	TCP	66	51144 → 5223 [ACK] Seq=257 Ack=3106 Win=2048

Frame 2984: Packet, 190 bytes on wire (1520 bits), 190 bytes captured (1520 bits) on interface 0

Ethernet II, Src: Apple 08:00:2e:14:61:00 (08:00:2e:14:61:00), Dst: zte 08:00:2e:14:61:00

Internet Protocol Version 4, Src: 192.168.1.5, Dst: 18.97.36.50

Transmission Control Protocol, Src Port: 51747, Dst Port: 443, Seq: 1816, Win: 126704, Len: 93

Transport Layer Security

DNS, TCP, UDP, TLSv1.3

Task 2:

Packet	Info Example	OSI Layer
DNS	"Standard query A <u>instagram.com</u> "	Application (Layer 7)
TCP	"443 → 51322 [SYN, ACK] Seq=0"	Transport (Layer 4)
TLS	Frame details showing source/destination IP	Network (Layer 3)

Task 3:

Application Layer Whatsapp / Instagram app interface
Presentation Layer Photo / Video encoding
Session Layer Keeping a video call session open
Transport Layer TCP splits message, ensures delivery
Network Layer IPs, Routing
Data Link Layer WiFi router, Ethernet frames
Physical Layer WiFi radio signals, cables, hardware

Task 4:

How a WhatsApp message travels through the OSI layers

- **Application layer** — WhatsApp (the app itself) takes your typed text and prepares it as data to be sent. This is where the actual "message" is created, before it becomes network traffic.
- **Presentation layer** — Your message gets encoded into a standard format and encrypted (WhatsApp uses end-to-end encryption), so it's unreadable to anyone intercepting it in transit.
- **Session layer** — A session is maintained between your phone and WhatsApp's servers so that your ongoing chat stays connected and in order, even if you send several messages in a row.
- **Transport layer** — Your encrypted message gets broken into smaller segments (if needed) and given sequence numbers, using TCP, so all pieces arrive reliably and in the correct order — and get reassembled correctly at the other end.
- **Network layer** — Each segment gets wrapped with IP addressing information — your phone's IP and WhatsApp's server IP — so routers along the way know where to send it.
- **Data link layer** — On your local network, the packet gets a MAC address wrapper so your Wi-Fi router knows exactly which device (your phone) it came from and where to send responses.
- **Physical layer** — The actual bits are converted into radio waves (Wi-Fi) or electrical/optical signals (if wired) and transmitted physically through the air or cables to your router and onward through the internet.

